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# Paleomagnetism and Potassium-Argon Ages of the Sonoma Volcanics, California

## ABSTRACT

Paleomagnetic data and potassium-argon ages indicate that the Sonoma Volcanics was erupted during the Pliocene Gilbert reversed and Gauss normal polarity epochs. The Gilbert reversed epoch is represented in the Howell Mountains east of Napa and east of St. Helena, in the mountains immediately east of the Valley of the Moon, and on the hill just north of Santa Rosa. The Gauss normal epoch is represented by the rocks from Mount St. Helena and possibly by the flows from the upper part of Sonoma Mountain and the rhyolite flows north of Sonoma. The age of the volcanic rocks ranges from 5.3 m.y. to about 2.9 m.y. Volcanic rocks approximately 11.8 m.y. old occur at Burdell Mountain northwest of Novato. This volcanic sequence is probably part of the Tolay Volcanics of the Petaluma area.

## INTRODUCTION

The Sonoma Volcanics crops out over an area of approximately 3,100 sq km in the Coast Ranges of California north of San Pablo Bay (Fig. 1). The maximum thickness of the sequence ranges between 300 and 600 m. The volcanic units consist of lava flows, tuffs, agglomerates, breccias, dikes, and plugs. The rock types vary from olivine basalt to andesite to silicic tuff and rhyolite. Intercalated sedimentary rocks containing volcanic debris are also found within the sequence.

The volcanic rocks were first described and subdivided by Osmont (1905). More detailed mapping showed that this sequence varies greatly from place to place, and Dickerson (1922) coined the name "Sonoma group" to include all the units. Morse and Bailey (1935) employed the name "Sonoma Volcanics," a term generally used since that time. Other reports include those by Bailey (1946), Yates and Hilpert (1946), Weaver (1949), Gealey (1951), Travis (1952), Cardwell (1958), and Kunkel and Upson (1960). Theses written about this area include those by Carter (1948), Goss (1948), Hurlbut (1948), Johnston (1948), and Crutchfield (1953). The general distribution of the Sonoma Volcanics is shown in the compilation by Koenig (1963), on a scale of 1:250,000 for the Geologic Map of California series, published by the California Division of Mines and Geology.

The Sonoma Volcanics has generally been assigned a middle or late Pliocene age because of its stratigraphic relation to sedimentary units of the area. Details of these relations have been described by Osmont (1905), Dickerson (1922), Weaver (1949), Travis (1952), and Cardwell (1958). A flora obtained from the Sonoma Volcanics was considered by Axelrod (1944) to be late Pliocene in age. A potassium-argon age of 3.4 m.y. for the tuff of the Petrified Forest southwest of Calistoga reported by Evernden and James (1964) also indicates a late Pliocene age.

The present paleomagnetic and geochronologic investigation resulted in a better determination of stratigraphic relations within the Sonoma Volcanics than was previously possible.

## POTASSIUM-ARGON DATING

Samples used for age determinations were collected from the same outcrops as those for paleomagnetic study. All feldspar concentrates were treated with a cold 10-percent solution of hydrofluoric acid for 10 to 20 min to remove impurities and to reduce the atmospheric argon contamination. Potassium was determined by flame photometry using an Instrumentation Laboratories flame photometer and chemical procedures described by Suhr and Ingamells (1966). Argon analyses were made by isotope dilution using procedures described by Dalrymple and Lanphere (1969). A bulb tracer with an approximate tracer size of $3 \times 10^{-10}$ mole $Ar^{38}$ per release was used. Mass analyses were made with a Reynolds-type mass spectrometer.

Three of the units (S14, S17, S20) gave age determinations with large errors (Table 1) be-

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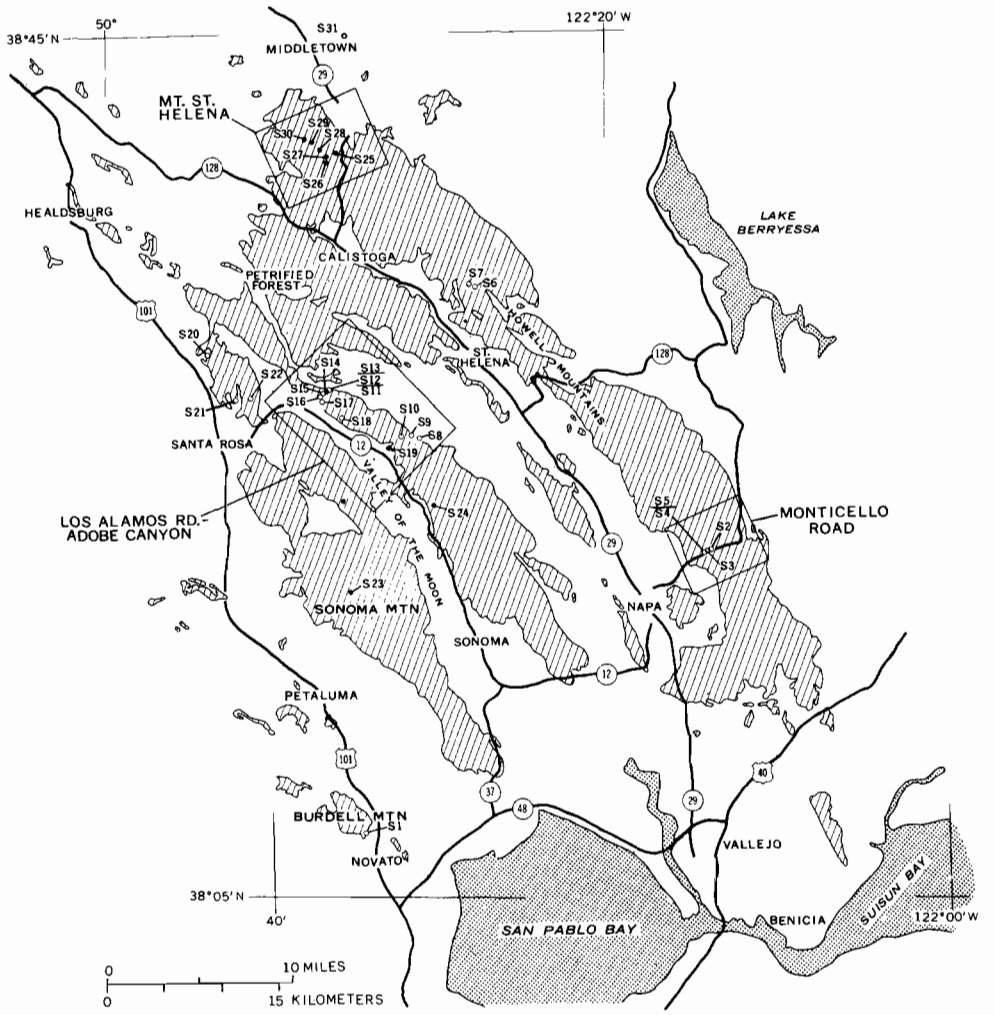



**Figure 1. Sample localities and the major occurrences of the Sonoma Volcanics (shaded). Solid symbols, normal polarity. Open symbols, reversed polarity. Outlined areas sampled in detail. Geologic data from Koenig (1963).**

cause of very low potassium contents in the plagioclase feldspars that led to large atmospheric argon corrections. X-ray diffractometer analyses of these feldspars showed their compositions to be in the labradorite to bytownite range. Petrographic determinations also indicate very calcic compositions approaching 90 percent anorthite.

## PALEOMAGNETIC STUDIES

Geographically oriented drill cores 2.5 cm in diameter were obtained from 18 separate rock units within the Sonoma Volcanics, using methods described by Doell and Cox (1967a). Five to nine cores were drilled from each unit over as great a lateral and vertical distance as possible to obtain a representative remanent magnetization direction for the entire flow. Oriented hand specimens, which were later cored in the laboratory, were collected from an additional 13 units. Specimens 2.3 cm long were cut from each core, and the natural remanent magnetization (NRM) of one specimen from each core was measured on a spinner magnetometer, described by Doell and Cox (1965). At least one specimen from each flow was progressively demagnetized using equip-

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TABLE 1. POTASSIUM-ARGON AGE DETERMINATIONS ON THE SONOMA VOLCANICS

| Unit | Material analyzed | $K_2O$ (wt. %) | Sample wt. (grams) | $Ar^{40}_{rad}$/gram ($10^{-12}$ moles) | $\frac{100\ Ar^{40}_{rad}}{Ar^{40}_{total}}$ | $\frac{Ar^{40}_{rad}}{K^{40}}$ ($10^{-4}$) | Calculated age* ($10^6$ years) |
|---|---|---|---|---|---|---|---|
| S1 | Plagioclase | 0.215<br>0.217 | 5.190 | 3.773 | 12.95 | 6.915 | 11.8 ± 0.8 |
| S2 | Plagioclase | 0.927<br>0.925 | 5.013 | 6.363 | 23.70 | 2.720 | 4.7 ± 0.2 |
| S5 | Plagioclase | 0.933<br>0.943 | 5.264 | 5.821 | 31.35 | 2.457 | 4.2 ± 0.1 |
| S8 | Whole rock | 3.75<br>3.77<br>3.75<br>3.77 | 9.426 | 29.24 | 49.77 | 3.078 | 5.3 ± 0.2 |
| S14 | Plagioclase | 0.0570<br>0.0603 | 5.848 | 0.3478 | 1.77 | 2.353 | 4.0 ± 1.2 |
| S17 | Plagioclase | 0.0622<br>0.0595 | 5.982 | 0.1290 | 0.92 | 0.8375 | 1.4 ± 0.8 |
| S20 | Plagioclase | 0.0443<br>0.0430 | 5.046 | 0.5712 | 1.96 | 5.198 | 8.9 ± 4.5 |
| S30 | Plagioclase | 1.625<br>1.640 | 5.048 | 7.066 | 14.96 | 1.713 | 2.9 ± 0.2 |

* $\lambda_\epsilon = 0.585 \times 10^{-10}$/yr, $\lambda_\beta = 4.72 \times 10^{-10}$/yr, $K^{40}/K = 1.19 \times 10^{-4}$ mole/mole. The (±) figures are estimates of precision at the 68-percent confidence level as calculated from Cox and Dalrymple (1967). See text for detailed discussion.

ment described by Doell and Cox (1967b) as a test of the stability of the remanent magnetization and as an indication of the approximate field strength to be used for magnetically cleaning the remaining samples from that lava.

The NRM directions for several of the units were tightly grouped, and partial demagnetization did not appreciably change their direction or improve their accuracy. For other units, magnetic cleaning at peak field strengths up to 400 oersteds was needed to determine the original thermoremanent magnetization (TRM) direction. For one unit (S30), it was not possible to remove completely all anomalous magnetizations and to determine the original TRM direction. However, when each specimen from this unit was progressively demagnetized, it became apparent that all samples tend toward an original normal direction (Fig. 2). The intensity of magnetization for all samples from this unit decreases very rapidly as is characteristic of flows affected by lightning, as described by Cox (1961) and Graham (1961).

The paleomagnetic data for all units studied are compiled in Table 2. The average magnetization directions after magnetic cleaning for all rock units sampled in this study are shown in Figure 3. For the units from which only hand samples were collected, the average direction was determined from two cores from each sample. For other units, the method developed by Fisher (1953) was used to determine the average direction.

Differences in magnetization directions between successive flows of the Mount St. Helena section (S25–S30), the Monticello Road section (S2–S5), and the Los Alamos Road–Adobe Canyon section (S8–S19), show sufficient changes in magnetization directions from secular variation to indicate that time intervals of a few hundred to a few thousand years were involved in their formation (Fig. 4). This is consistent with the great thickness of the Mount St. Helena and

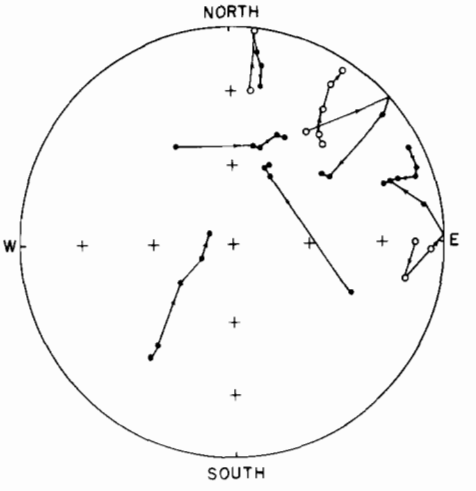



**Figure 2. Changes in magnetic directions of all samples from unit S30 during demagnetization. All samples demagnetized to peak fields of 1,000 oersteds. Open symbols represent directions on the upper hemisphere. Solid symbols, directions on the lower hemisphere. Equal-area projection.**

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TABLE 2. PALEOMAGNETIC DATA

| Unit | Site locality lat | long | N | H (oe) | RM direction Incl. | Decl. | R | K | $\alpha_{95}$ | J x $10^{-4}$ ± s.d. | VGP position PLA | PLO |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 38.13 | 237.43 | 8 | 100 | -71.1 | 277.1 | 7.91623 | 84 | 6.1 | 0.06 ± 0.0 | -27.1 | 96.5 |
| 2 | 38.35 | 237.77 | 8 | NRM | -35.5 | 177.1 | 7.99457 | 1,287 | 1.6 | 0.68 ± 0.2 | -71.1 | 246.2 |
| 3 | 38.35 | 237.77 | 8 | 200 | -54.6 | 188.1 | 7.96093 | 179 | 4.2 | 16.1 ± 21. | -82.8 | 171.9 |
| 4 | 38.35 | 237.77 | 8 | 200 | -47.0 | 182.7 | 7.93587 | 109 | 5.3 | 0.83 ± 0.4 | -79.6 | 224.6 |
| 5 | 38.35 | 237.77 | 8 | 100 | -48.3 | 181.2 | 7.95952 | 173 | 4.2 | 5.06 ± 5.8 | -80.9 | 231.5 |
| 6 | 38.56 | 237.53 | 8 | NRM | -31.1 | 185.8 | 7.98117 | 372 | 2.9 | 5.39 ± 2.0 | -67.6 | 222.9 |
| 7 | 38.56 | 237.53 | 8 | 200 | -68.2 | 166.4 | 7.99228 | 906 | 1.9 | 7.94 ± 1.9 | -74.1 | 25.1 |
| 8 | 38.44 | 237.48 | 2 | 100 | -53. | 172. | | | | 0.05 | | |
| 9 | 38.44 | 237.47 | 2 | 200 | -30. | 169. | | | | 9.59 | | |
| 10 | 38.44 | 237.46 | 2 | NRM | Reversed(?) | | | | | 0.003 | | |
| 11 | 38.48 | 237.39 | 8 | 200 | -40.6 | 200.3 | 7.91276 | 80 | 6.2 | 4.71 ± 2.1 | -66.9 | 182.9 |
| 12 | 38.48 | 237.39 | 8 | NRM | -34.2 | 182.1 | 7.97530 | 283 | 3.3 | 57.4 ± 15. | -70.2 | 231.6 |
| 13 | 38.48 | 237.38 | 2 | 200 | -24. | 174. | | | | 5.01 | | |
| 14 | 38.48 | 237.38 | 6 | NRM | +50.2 | 345.0 | 5.98345 | 302 | 3.9 | 13.1 ± 0.7 | 75.6 | 120.6 |
| 15 | 38.47 | 237.38 | 7 | 200 | -40.2 | 131.2 | 6.93583 | 94 | 6.3 | 5.87 ± 4.0 | -45.8 | 321.4 |
| 16 | 38.47 | 237.38 | 8 | NRM | -26.1 | 176.3 | 7.98681 | 530 | 2.4 | 10.4 ± 1.5 | -65.1 | 246.0 |
| 17 | 38.47 | 237.38 | 2 | NRM | -52. | 175. | | | | 3.06 | | |
| 18 | 38.46 | 237.40 | 2 | NRM | -40. | 200. | | | | 31.5 | | |
| 19 | 38.43 | 237.45 | 9 | NRM | +61.7 | 341.0 | 8.95525 | 179 | 3.9 | 59.6 ± 40. | 74.9 | 170.7 |
| 20 | 38.50 | 237.27 | 2 | NRM | -57. | 223. | | | | 19.9 | | |
| 21 | 38.47 | 237.30 | 8 | 100 | -45.1 | 161.8 | 7.83382 | 42 | 8.6 | 2.20 ± 0.5 | -70.7 | 294.9 |
| 22 | 38.47 | 237.31 | 2 | NRM | -44. | 206. | | | | 35.0 | | |
| 23 | 38.32 | 237.41 | 2 | 200 | +54. | 20. | | | | 0.97 | | |
| 24 | 38.38 | 237.49 | 2 | 50 | +24. | 303. | | | | 0.20 | | |
| 25 | 38.66 | 237.40 | 5 | 400 | +21.1 | 303.4 | 4.86785 | 30 | 14. | 1.12 ± 1.7 | 32.7 | 134.3 |
| 26 | 38.65 | 237.39 | 2 | 200 | +58. | 318. | | | | 0.02 | | |
| 27 | 38.65 | 237.39 | 2 | 200 | +28. | 1. | | | | 0.07 | | |
| 28 | 38.66 | 237.38 | 5 | 400 | +50.8 | 0.0 | 4.95681 | 93 | 8.0 | 1.15 ± 0.3 | 82.8 | 57.4 |
| 29 | 38.67 | 237.38 | 2 | NRM | +51. | 7. | | | | 8.21 | | |
| 30 | 38.67 | 237.37 | 8 | 1000 | +37.1 | 40.7 | 6.72252 | 6 | 26. | 0.78 ± 0.3 | 50.8 | 342.7 |
| 31 | 38.75 | 237.41 | 8 | 100 | -51.0 | 178.3 | 7.98392 | 435 | 2.7 | 11.9 ± 2.1 | -82.8 | 249.1 |

*N = Number of samples used in analysis.*
*H = Peak field strength used in magnetic cleaning.*
*RM = Remanent magnetization. I, positive downward, D, eastward.*
*R = Vector sum of N unit vectors.*
*K = (N-I)/(N-R); (Fisher, 1953).*
$\alpha_{95}$ *= 95-percent confidence cone about average direction (Fisher, 1953).*
*J = Intensity of magnetization after magnetic cleaning in emu/cc.*
*PLA, PLO = North latitude and east longitude, respectively, of the VGP position (Cox and Doell, 1960).*

Los Alamos Road–Adobe Canyon sections and the numerous interbeds of tuffaceous sediments.

The section of volcanic rocks exposed on Mount St. Helena may be the youngest volcanic section exposed in the area. This section consists of approximately 600 m of flows and densely welded tuff of rhyolitic composition intercalated with poorly consolidated tuff. These rocks overlie the thick tuff sequence that was originally named the "Sonoma Tuff" by Osmont (1905). All units sampled in this section have a normal polarity. The lowest unit sampled, S25, is a breccia that appears to occupy a small vent at the base of the rhyolite and is younger than the thick underlying sequence of tuff. This vent may have served as a feeder for one of the overlying rhyolite flows.

Several andesitic welded tuff units are exposed along Monticello Road, and those sampled have a reversed polarity. The two youngest units, S4 and S5, were probably formed in a brief time interval, since there is no significant difference between their magnetization directions. More than 30 m of unconsolidated tuff occurs between these units and S2, the oldest welded tuff unit sampled. Unit S2 has been cut by a thick dike, S3, that has a magnetization direction similar to the direction of the younger tuffs, indicating a comparable age.

Numerous basalt, andesite, and rhyolite flows intercalated with tuffs and tuffaceous sediments are exposed in the Los Alamos Road–Adobe Canyon section. All the units except S14 and S19 have a reversed polarity. S14 is

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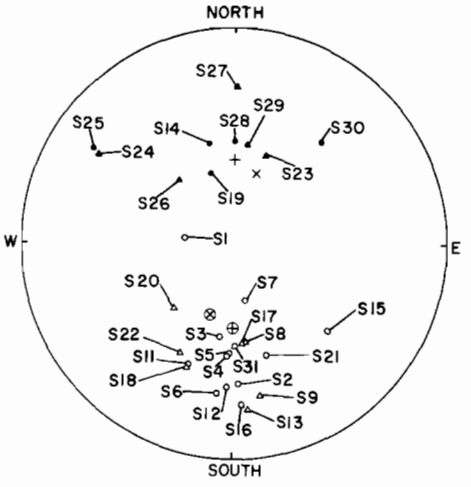



Figure 3. Average remanent magnetization directions for all rock units after magnetic cleaning. Circles are units where multiple cores were collected. Triangles represent hand samples. Open symbols represent directions on the upper hemisphere. Solid symbols, directions on the lower hemisphere. +, ideal dipole field direction; X, present field direction. Equal-area projection.

within the reversed part of the sequence and may represent a short polarity event. S19 occurs at the very top of the sequence, and it is not possible to determine if this unit represents another polarity event or the beginning of a normal polarity epoch. Unit S10 is questionably reversely magnetized. While the NRM indicates a reversed direction, this unit is too weakly magnetized for partial demagnetization, so that the reliability of its direction could not be determined.

Virtual geomagnetic poles (VGP) were calculated for all units drilled in the field using the method developed by Cox and Doell (1960), and these are shown in Figure 5. A mean field direction and a mean of the VGP directions were calculated (Table 3) for those units whose directions were determined with considerable accuracy ($\alpha_{95} < 10°$), except units S1 and S31. S1 is from an older volcanic sequence and S31 is younger than the Sonoma Volcanics. These data show that the geomagnetic field at the time of formation of the Sonoma Volcanics was not very different from an axial geocentric dipole field.

## DISCUSSION

The stratigraphy, magnetic polarity, and calculated ages of the dated sequences are

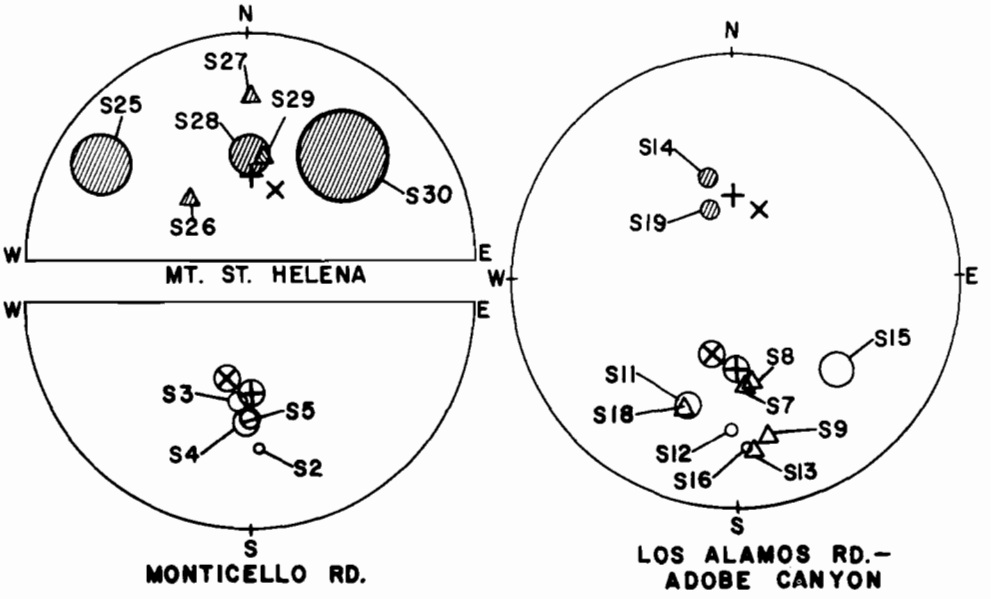



Figure 4. Average remanent magnetization directions of units within the Sonoma Volcanics from the Mount St. Helena, Monticello Road, and Los Alamos Road–Adobe Canyon sections. Circles are confidence limits about the average directions. Triangles show an $\alpha_{95}$ about 5°. Symbols same as for Figure 3.

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TABLE 3. FLOW AVERAGE DATA

| Data | No. of units | Incl. | Decl. | R | K | $\alpha_{95}$ |
|---|---|---|---|---|---|---|
| Field directions | 14 | +46.4 | 355.1 | 13.4645 | 24.0 | 8.2 |
| | | lat | long | | | |
| VGP's | 14 | 79.4N. | 83.3E. | 13.3780 | 21.0 | 8.9 |

shown in Figure 6 in relation to the polarity time scale known from previous studies and inferred from mid-oceanic magnetic anomalies.

The youngest unit, S31, is part of a group of basalt flows that overlie erosional remnants of the Sonoma Volcanics. Because of the relation of these basalt flows to present topography, Yates and Hilpert (1946) favor a Pleistocene age for their formation. The reversed polarity of unit S31 places its formation during the Matuyama reversed polarity epoch which began 2.4 m.y. ago and ended 0.70 m.y. ago.

The thick sequence of normally magnetized rhyolite units forming Mount St. Helena was deposited during the early part of the Gauss normal polarity epoch, as the age of 2.9 m.y. for the uppermost unit indicates. As no rocks that were erupted during the Mammoth reversed polarity event at 3.06 m.y. were sampled during this study, it is not known if they are represented in this sequence. The unique magnetization directions for units S24 and S25 suggest that they may have formed at about the same time and indicate a possible correlation between some of the rhyolite flows near Sonoma with those of Mount St. Helena. The normal unit S23 may also indicate that the flows from the upper part of Sonoma Mountain belong in this period of volcanism.

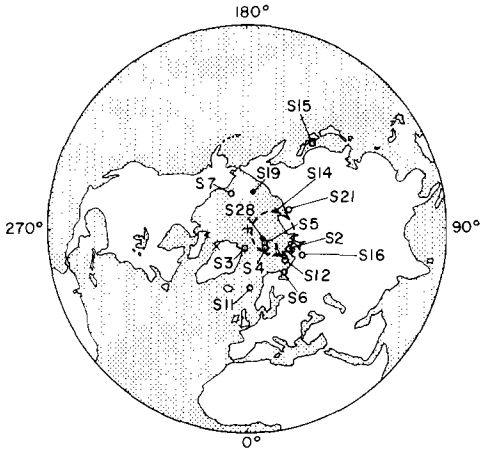



**Figure 5. Virtual geomagnetic poles (VGP) for the Sonoma Volcanics. Solid circles, normal polarity. Open circles, projection of reversed VGP's on the upper hemisphere. Dotted circle defines the 95-percent confidence limits about the average direction for these units. +, ideal dipole field; X, VGP calculated from present field direction.**

The reversed section of welded tuff units along Monticello Road east of Napa was formed during the Gilbert reversed polarity epoch before the Nunivak normal polarity event that has been dated between 4.04 and 4.14 m.y. The normal polarity event at about 4.4 m.y. suggested by the mid-oceanic anomalies reported by Heirtzler and others (1968) was not represented in this section. The reversely magnetized rock units S6 and S7 northeast of St. Helena are probably part of the sequence seen in the Monticello Road section.

The sequence of lavas in the Los Alamos Road–Adobe Canyon section east of the Valley of the Moon was also formed during the Gilbert reversed polarity epoch as indicated by the age of 5.3 m.y. for the lowest unit in the sequence. The position of the upper part of this sequence is in doubt because of the unreliable ages obtained for units S14 and S17. One interpretation is that the normal units, S14 and S19, represent the normal events at about 4.9 and 4.7 m.y., suggested by Heirtzler and others (1968). However, polarity events are often of such short duration that all events are not necessarily represented in any one volcanic sequence. Also, the thickness of the reversed section of andesite flows between these two units makes it unlikely that they could have formed in the short time interval between these two events. R. L. Rose (1970, oral commun.) reports a suspected hiatus between units S10 and S14, suggesting a younger age for the upper part that may make it similar in age to the Monticello Road section. It would be possible to assign a minimum age for this part of the section if the normal unit, S19, represents the beginning of the Gauss normal polarity epoch at 3.32 m.y., and if unit S14 is correlative to the Cochiti normal polarity event at 3.67 m.y. The reversely magnetized rock units S20, S21, and S22, immediately north of Santa Rosa, are probably correlative with part of the Los Alamos Road–Adobe Canyon section.

The reversed flow near the base of Burdell Mountain has a K-Ar age of 11.8 m.y., so this unit should have formed during a short normal polarity event suggested by the mid-oceanic

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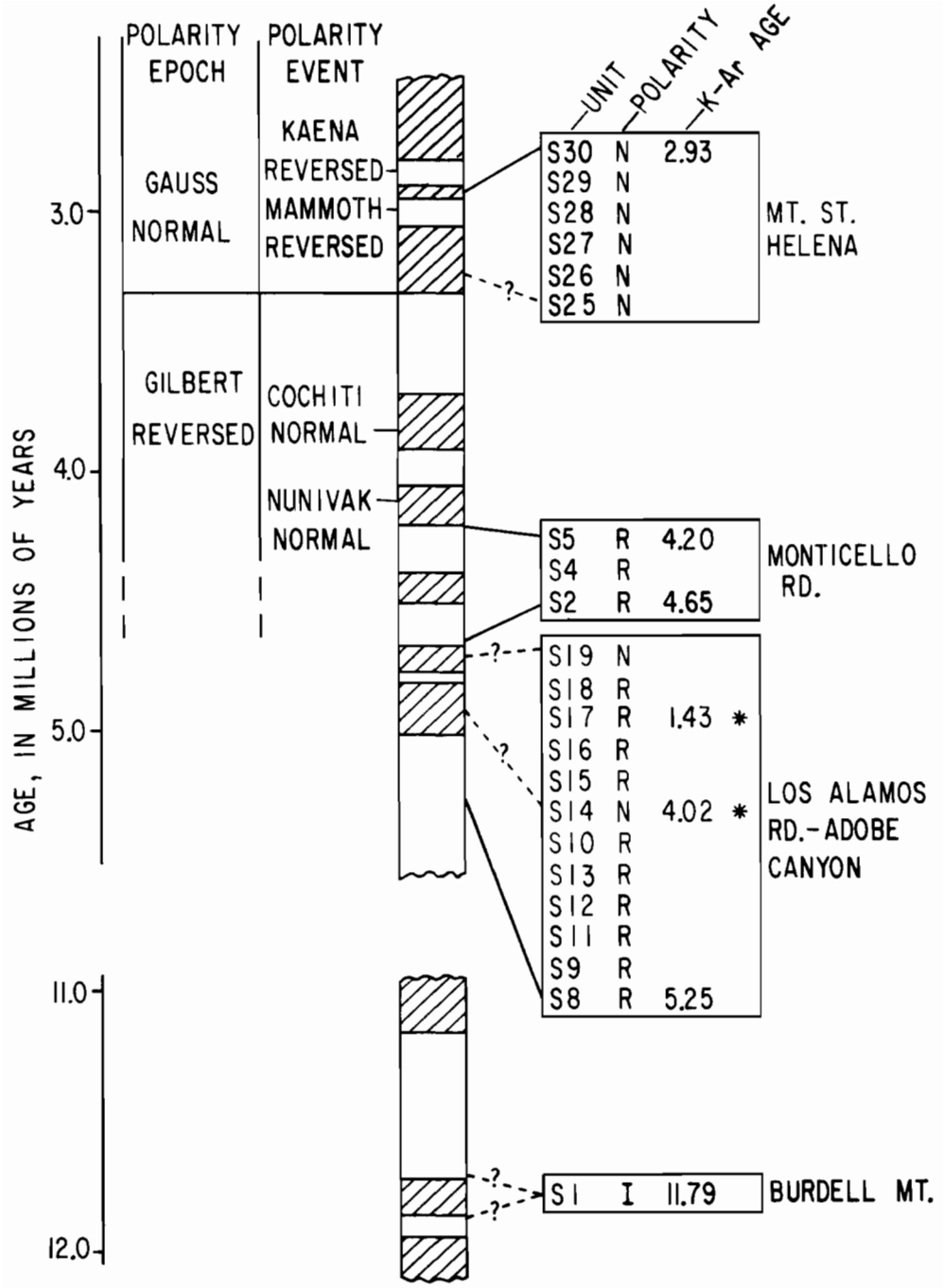



Figure 6. Relations of stratigraphy, magnetic polarity, and K-Ar ages of the Sonoma Volcanics to the reversal time scale. Shaded parts of scale represent periods of normal polarity. Polarities: N, normal; R, reversed; I, intermediate. * denotes ages unreliable for reasons discussed in text. Time scale constructed using data from Cox (1969) and Heirtzler and others (1968).

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anomalies. This apparent inconsistency is not serious, since the imprecision of the date could place the unit at either side of the event. The unusual magnetization direction suggests that the unit may have formed during a polarity transition. Although most geologic reports of the area assign the volcanic rocks of Burdell Mountain to the Sonoma Volcanics, the present study shows that they are early Pliocene and are probably correlative with the Tolay Volcanics of the Petaluma oil field described by Morse and Bailey (1935).

## ACKNOWLEDGMENTS

I wish to thank R. R. Doell and G. B. Dalrymple for their advice and encouragement during the course of this study. Special thanks are also due to R. L. Rose of San Jose State College, whose knowledge of the Sonoma Volcanics was very helpful. Potassium analyses were made by Lois Schlocker. This work forms part of a thesis submitted to San Jose State College, California, in partial satisfaction of the requirements for a Master of Science degree.

## APPENDIX. SAMPLE LOCALITIES

All maps listed are U.S. Geological Survey 7½ min topographic quadrangle maps, scale 1:24,000. Flow numbers are those used in text. Numbers in parentheses are field numbers.

**S1 (9E001).** lat 38.13° N., long 237.43° E. Lowest andesite flow on Burdell Mountain. Sampled at quarry on light-duty road across U.S. Highway 101 from Basalt Creek. Eight cores. South center Petaluma River quadrangle.

**S2 (9E048).** lat 38.35° N., long 237.77° E. Attitude N. 70° E., +15° NW. Oldest welded tuff flow sampled along Monticello Road. Approximately 1 km southwest of BM 973. Eight cores. Northwest corner Mount George quadrangle.

**S3 (9E056).** lat 38.35° N., long 237.77° E. Large dike 30 m west of and cutting unit S2. Eight cores. Monticello Road, Mount George quadrangle.

**S4 (8E033).** lat 38.35° N., long 237.77° E. Attitude N. 70° E., +15° NW. Welded tuff flow on east side of dirt road joining Monticello Road at elevation 626. Eight cores. Mount George quadrangle.

**S5 (8E025).** lat 38.35° N., long 237.77° E. Attitude N. 70° E., +15° NW. Youngest welded tuff sampled along Monticello Road. Stratigraphically above and just across dirt road from S4. Eight cores. Mount George quadrangle.

**S6 (9E064).** lat 38.56° N., long 237.53° E. Attitude N. 70° W., +20° SW. Welded tuff unit along Deer Park Road approximately 1.1 km northwest of BM 1394 at Four Corners intersection. Eight cores. West center section of St. Helena quadrangle.

**S7 (9E072).** lat 38.56° N., long 237.53° E. Attitude N. 70° W., +20° SW. Andesite flow stratigraphically above and approximately 0.4 km northwest of S6 along Deer Park Road. Eight cores. St. Helena quadrangle.

**S8 (9E119).** lat 38.44° N., long 237.48° E. Attitude N. 50° W., +75° SW. Rhyolite flow forming base of volcanics in this section. Adobe Canyon Road approximately 0.6 km east of bridge crossing Sonoma Creek. Hand sample. East center Kenwood quadrangle.

**S9 (9E120).** lat 38.44° N., long 237.47° E. Approximately attitude N. 50° W., +35° SW. Rhyolite flow along Adobe Canyon Road approximately 9 m west of bridge crossing Sonoma Creek. Hand sample. East center Kenwood quadrangle.

**S10 (9E099).** lat 38.44° N., long 237.46° E. Attitude N. 50° W., +30° SW. Rhyolite flow approximately 0.3 km northwest of road junction at elevation 578. Hand sample. Adobe Canyon Road. Center Kenwood quadrangle.

**S11 (8E009).** lat 38.48° N., long 237.39° E. Attitude N. 80° W., +80° SW. Oldest andesite flow exposed along Los Alamos Road. Approximately 0.5 km SW of water tank. Eight cores. Northwest corner Kenwood quadrangle.

**S12 (8E017).** lat 38.48° N., long 237.39° E. Attitude N. 80° W., +50° SW. Andesite flow stratigraphically above and 15 m west of S11 on Los Alamos Road. Eight cores. Kenwood quadrangle.

**S13 (9E121).** lat 38.48° N., long 237.38° E. Attitude N. 80° W., +50° SW. Andesite flow stratigraphically above and 12 m south of S12 on Los Alamos Road. Hand sample. Kenwood quadrangle.

**S14 (9E111).** lat 38.48° N., long 237.38° E. Attitude N. 60° W., +50° SW. Andesite flow stratigraphically above and 12 m southwest of S13 on Los Alamos Road. Six cores. Kenwood quadrangle.

**S15 (8E001).** lat 38.47° N., long 237.38° E. Attitude N. 65° E., +40° SW. Andesite flow stratigraphically above S14. Nine m northeast of quarry on Los Alamos Road. Seven cores. Kenwood quadrangle.

**S16 (9E010).** lat 38.47° N., long 237.38° E. Attitude N. 20° W., +30° SW. Youngest andesite flow exposed on Los Alamos Road. Approximately 0.2 km northeast of road junction at elevation 573. Eight cores. Kenwood quadrangle.

**S17 (9E122).** lat 38.47° N., long 237.38° E. Attitude N. 40° W., +30° SW. Basalt flow on Wildwood Terrace stratigraphically above S16. Approximately 0.5 km from intersection of Wildwood Terrace and Los Alamos Road. Hand sample. Kenwood quadrangle.

**S18 (9E123).** lat 38.46° N., long 237.40° E. Approximately attitude N. 40° W., +20° SW. Andesite flow stratigraphically above S17 and below S19. Unit forms hill 1061 south of Buzzard Peak. Approximately 1.3 km along dirt road 2.7 km north-

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west of Pythian Road along Highway 12. Hand sample. West center Kenwood quadrangle.

**S19 (9E101).** lat 38.43° N., long 237.45° E. Attitude N. 50° W., +20° SW. Andesite flow. Youngest flow sampled along Los Alamos Road–Adobe Canyon section. Approximately 1.0 km from State Highway 12 along Adobe Canyon Road. Eight cores. Kenwood quadrangle.

**S20 (9E090).** lat 38.50° N., long 237.27° E. Basalt flow on Mark West Springs Road. 0.3 km northeast of BM 184. Hand sample. Southwest corner Mark West Springs quadrangle.

**S21 (9E080).** lat 38.47° N., long 237.30° E. Basalt flow near County Hospital on Chanate Road. 60 m north of entrance to Hillcrest Detention Home. Eight cores. Center Santa Rosa quadrangle.

**S22 (9E088).** lat 38.47° N., long 237.31° E. Basalt flow along Chanate Road. 0.3 km west of intersection with Montecito Road. Hand sample. Center Santa Rosa quadrangle.

**S23 (9E009).** lat 38.32° N., long 237.41° E. Attitude N. 10° E., +35° SE. Andesite flow on Sonoma Mountain Road northeast of Petaluma at 1,440 ft elev. Approximately 1.4 km north of BM 1246. Hand sample. West center Glen Ellen quadrangle.

**S24 (9E110).** lat 38.38° N., long 237.49° E. Banded rhyolite flow at Trinity Quarry. Quarry along dirt road joining Trinity Road approximately 0.3 km east of State Highway 12. Hand sample. Southwest corner Kenwood quadrangle.

**S25 (9E043).** lat 38.66° N., long 237.40° E. Vent breccia. Lowest unit sampled on Mount St. Helena. Along road to top of Mount St. Helena at 2,480 ft elev. Five cores. West center Detert Reservoir quadrangle.

**S26 (9E042).** lat 38.65° N., long 237.39° E. Rhyolite flow. Road to top of Mount St. Helena at 2,960 ft elev. Hand sample. Detert Reservoir quadrangle.

**S27 (9E041).** lat 38.65° N., long 237.39° E. Rhyolite flow. Road to top of Mount St. Helena at 3,300 ft elev. Hand sample. Detert Reservoir quadrangle.

**S28 (9E035).** lat 38.66° N., long 237.38° E. Rhyolite flow. Road to top of Mount St. Helena at 3,820 ft elev. 60 m below road leading to radio relay station on south peak. Five cores. Detert Reservoir quadrangle.

**S29 (9E034).** lat 38.67° N., long 237.38° E. Rhyolite flow. Road to top of Mount St. Helena at 4,040 ft elev. Hand sample. Detert Reservoir quadrangle.

**S30 (9E021).** lat 38.67° N., long 237.37° E. Rhyolite flow at top of Mount St. Helena. Sampled at VABM 4343. Eight cores. Center east edge of Mount St. Helena quadrangle.

**S31 (9E018).** lat 38.75° N., long 237.41° E. Basalt flow of probable Pleistocene age. Exposed on west side of Butts Canyon Road approximately 6 m north of intersection of road with south edge of map. Eight cores. Middletown quadrangle.

## REFERENCES CITED


Axelrod, D. I., 1944, The Sonoma flora, *in* Chaney, R. W., ed., Pliocene floras of California and Oregon: Carnegie Inst. Washington Pub. 553, p. 167–206.

Bailey, E. H., 1946, Quicksilver deposits of the western Mayacmas district, Sonoma County, California: California Div. Mines Rept. 42, no. 3, p. 199–230.

Cardwell, G. T., 1958, Geology and groundwater in the Santa Rosa and Petaluma Valley areas, Sonoma County, California: U.S. Geol. Survey Water-Supply Paper 1427, 273 p.

Carter, W. H., 1948, Geology of the northeast corner of the Calistoga quadrangle [M.S. thesis]: Berkeley, Univ. California, 72 p.

Cox, Allan, 1961, Anomalous remanent magnetization of basalt: U.S. Geol. Survey Bull. 1083-E, p. 131–160.

—— 1969, Geomagnetic reversals: Science, v. 163, p. 237–245.

Cox, Allan, and Doell, R. R., 1960, Review of paleomagnetism: Geol. Soc. America Bull., v. 71, p. 645–768.

Cox, Allan, and Dalrymple, G. B., 1967, Statistical analysis of geomagnetic reversal data and the precision of potassium-argon dating: Jour. Geophys. Research, v. 72, p. 2603–2614.

Crutchfield, W. H., Jr., 1953, The geology and silver mineralization of the Calistoga district, Napa County, California [M.A. thesis]: Berkeley, Univ. California, 71 p.

Dalrymple, G. B., and Lanphere, M. A., 1969, Potassium-argon dating: Principles, techniques, and applications to geochronology: San Francisco, W. H. Freeman and Co., 258 p.

Dickerson, R. E., 1922, Tertiary and Quaternary history of the Petaluma, Point Reyes, and Santa Rosa quadrangles: California Acad. Sci., 4th ser., v. 11, no. 19, p. 527–601.

Doell, R. R., and Cox, Allan, 1965, Measurement of the remanent magnetization of igneous rocks: U.S. Geol. Survey Bull. 1203-A, p. 1–32.

—— 1967a, Paleomagnetic sampling with a portable coring drill, *in* Collinson, D. W., Creer, K. M., and Runcorn, S. K., eds., Methods in paleomagnetism: Amsterdam, Elsevier Pub. Co., p. 21–25.

—— 1967b, Analysis of alternating field demagnetization equipment, *in* Collinson, D. W., Creer, K. M., and Runcorn, S. K., eds., Methods in paleomagnetism: Amsterdam, Elsevier Pub. Co., p. 241–253.

Evernden, J. F., and James, G. T., 1964, Potassium-argon dates and the Tertiary floras of North America: Am. Jour. Sci., v. 262, p. 945–974.

Fisher, R. A., 1953, Dispersion on a sphere: Royal Astron. Soc. London Proc., ser. A, v. 217, p. 295–305.

Gealey, W. K., 1951, Geology of the Healdsburg

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quadrangle, California: California Div. Mines Bull. 161, p. 1–50.

Goss, C. R., 1948, Geology of the southwest corner of the Calistoga quadrangle [M.A. thesis]: Berkeley, Univ. California, 45 p.

Graham, K. W. T., 1961, The re-magnetization of a surface outcrop by lightning currents: Royal Astron. Soc. Geophys. Jour., v. 6, p. 85–102.

Heirtzler, J. R., Dickson, G. O., Herron, E. M., Pitman, W. C., III, and LePichon, X., 1968, Marine magnetic anomalies, geomagnetic field reversals, and motions of the ocean floor and continents: Jour. Geophys. Research, v. 73, p. 2119–2136.

Hurlbut, E. M., 1948, Geology of a portion of the Calistoga quadrangle [M.A. thesis]: Berkeley, Univ. California, 53 p.

Johnston, Stedwell, 1948, The geology of a portion of the Calistoga quadrangle, California [M.A. thesis]: Berkeley, Univ. California, 47 p.

Koenig, J. B., 1963, Santa Rosa sheet: Geologic map of California: California Div. Mines and Geology, scale 1:250,000.

Kunkel, Fred, and Upson, J. E., 1960, Geology and groundwater in Napa and Sonoma Valleys, Napa and Sonoma Counties, California: U.S. Geol. Survey Water-Supply Paper 1495, 252 p.

Morse, R. R., and Bailey, T. L., 1935, Geological observations in the Petaluma district, California: Geol. Soc. America Bull., v. 46, p. 1437–1456.

Osmont, V. C., 1905, A geological section of the Coast Ranges north of the Bay of San Francisco: California Univ. Dept. Geol. Sci. Bull., v. 4, p. 39–87.

Suhr, N. H., and Ingamells, C. O., 1966, Solution technique for analysis of silicates: Anal. Chemistry, v. 38, p. 730–734.

Travis, R. B., 1952, Geology of the Sebastapol quadrangle, California: California Div. Mines Bull. 162, 33 p.

Weaver, C. E., 1949, Geology of the Coast Ranges immediately north of the San Francisco Bay region, California: Geol. Soc. America Mem. 35, 242 p.

Yates, R. G., and Hilpert, L. S., 1946, Quicksilver deposits of the eastern Mayacmas district, Lake and Napa Counties, California: California Div. Mines Rept. 42, no. 3, p. 231–286.

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